

HOW TO RUN THE AI HEALTH COMPANION PROTOTYPE

This guide explains how to run the project on a new system or laptop.

1. Install Required Software

Before running the project, make sure the following are installed:

- Python 3.11 or above
- VS Code or any Python-supported IDE
- Internet connection (required for AI chat features)

2. Copy or Download the Project

Copy the complete project folder:

HYBRID-HEALTH-RISK-SYSTEM

to your system.

You can:

- clone it from GitHub ([Link to Github repo](#))
- download ZIP and extract it
- copy the folder manually

3. Open the Project

Open the project folder in:

VS Code

or any preferred IDE.

4. Create a Virtual Environment

Open terminal inside the project folder and run:

```
python -m venv venv
```

This creates a virtual environment for the project.

5. Activate the Virtual Environment (through terminal)

For Windows

```
venv\Scripts\activate
```

For Mac/Linux

```
source venv/bin/activate
```

After activation, (venv) will appear in the terminal.

6. Install Required Libraries

Run the following command:

```
pip install -r requirements.txt
```

This installs all required dependencies for the project.

7. Configure Environment Variables

Create a file named:

```
.env
```

inside the root folder.

Add the following variables:

```
OPENROUTER_API_KEY=your_api_key
```

```
SECRET_KEY=y7f9e2b4c6d1a8e0f3c5b9d2a7f4e6c1b8a0d3f5c9e2b7a1d6f4c8e0b2a9d5f1
```

Replace the placeholder values with actual keys.

⚠ Important Configuration Note

Inside the .env file, keep the SECRET_KEY exactly as provided and copy-paste it directly without making any changes.

You only need to configure these two variables inside the .env file:

```
OPENROUTER_API_KEY=your_openrouter_api_key
```

```
SECRET_KEY= y7f9e2b4c6d1a8e0f3c5b9d2a7f4e6c1b8a0d3f5c9e2b7a1d6f4c8e0b2a9d5f1
```

The OPENROUTER_API_KEY must be generated manually by following the steps provided below.

No other variables are required inside the .env file for running this prototype.

7.1 Generating the OpenRouter API Key

The project uses OpenRouter for:

- AI wellness interpretation
- conversational AI assistant
- personalized AI responses

Follow these steps to generate the API key:

Step 1

Visit the official OpenRouter website:

[OpenRouter Official Website](#)

Step 2

Create an account or login using:

- Google account
- GitHub account
- Email login

Step 3

After login, open the API Keys page:

[OpenRouter API Keys Page](#)

Step 4

Click:

“Create Key”

and generate a new API key.

Step 5

Copy the generated API key and paste it inside the .env file:

```
OPENROUTER_API_KEY=sk-or-v1-xxxxxxxxxxxxxxxxxxxx
```

⚠ Important Notes

- Do NOT share your API key publicly
- Do NOT upload the .env file to GitHub
- Keep all keys private and secure

The AI chat and wellness interpretation features will not work without a valid OpenRouter API key.

8. Start the FastAPI Backend Server

Run the following command:

```
uvicorn project.api.server:app --reload
```

The backend server will start at:

<http://127.0.0.1:8000>

Keep this terminal running.

9. Start the Streamlit Frontend

Open a new terminal inside the project folder.

Activate the virtual environment again and run:

```
streamlit run project/ui/app.py
```

The frontend dashboard will open automatically in the browser.

Usually at:

<http://localhost:8501>

10. Using the Prototype

After the application starts, users can:

- create an account
- login to the system
- enter health information
- generate AI health assessments
- view wellness analytics
- chat with the AI wellness assistant
- download PDF reports
- access previous reports

11. Important Notes

AI Disclaimer

This prototype is:

NOT a medical diagnostic system.

The generated predictions are AI-based estimations intended only for:

- educational purposes
- research demonstrations
- preventive wellness awareness

Internet Requirement

Internet connection is required for:

- AI wellness interpretation
- conversational AI assistant

because they depend on external AI APIs.

12. Common Issues & Solutions

Problem	Solution
Libraries not installed	Run <code>pip install -r requirements.txt</code> again
Backend not connecting	Ensure FastAPI server is running
AI responses not working	Check API keys inside <code>.env</code>
PDF export issue	Ensure ReportLab is installed properly
Port already in use	Restart terminal or change port

13. Stopping the Application

To stop the application, press:

CTRL + C

inside both terminal windows:

- FastAPI terminal
- Streamlit terminal

14. Final Note

The project is designed as a modular AI healthcare prototype demonstrating:

- Machine Learning integration
- Deep Learning inference
- Conversational AI
- wellness analytics
- modern healthcare dashboard systems

within a single full-stack application.